

Graphene Nanosheets

Technical Data Sheet

*Unlocking Graphene's
Full Potential*



Graphene Nanosheets

➤ Supply Form - Powder

Our proprietary microwave plasma technology enables the continuous, one-step synthesis of high-quality graphene.

The technology offers unprecedented control over the material structure and composition, producing graphene with an oxygen content below 1%, a high sp^2/sp^3 carbon ratio, and nearly 40% single-layer graphene sheets with lateral dimensions of around $0.5\ \mu m$.

The synthesized materials exhibit a high specific surface area and excellent electrical conductivity exceeding $3000\ S\ m^{-1}$, making them ideally suited for demanding applications in energy storage, electronics, electromagnetic shielding, sensors, catalysis, and advanced composite materials.

Graphene Nanosheets Characterization	Value
Thickness [nm]	≤ 1
Average size [nm]	~450
Oxygen [%]	~ 1
sp^3 carbons [%]	< 10
Conductivity [S/m]	> 3500
Specific Surface area [m ² /g]	> 400
Monolayer sheets [%]	> 40

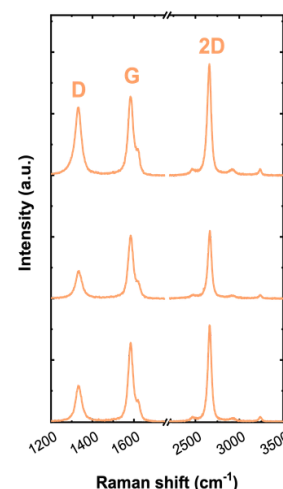
Graphene Characterization & Techniques

Raman Spectroscopy

LabRAM HR Visible (Horiba Jobin-Yvon) Raman spectrometer at 633 nm, with 5 cm⁻¹ spectral resolution and a laser spot-size of 2mm.

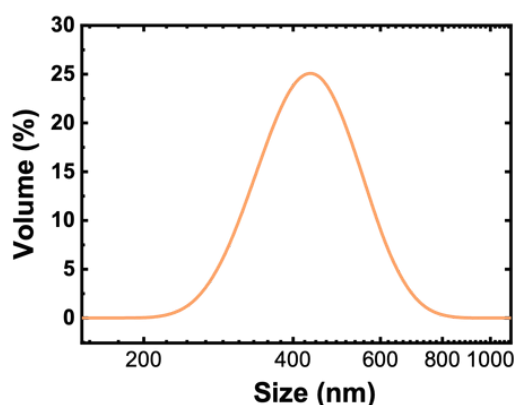
Measurements were performed with a laser power $P = 0.054$ mW to avoid overheating.

Raman spectra are collected from different locations of the sample under analysis.

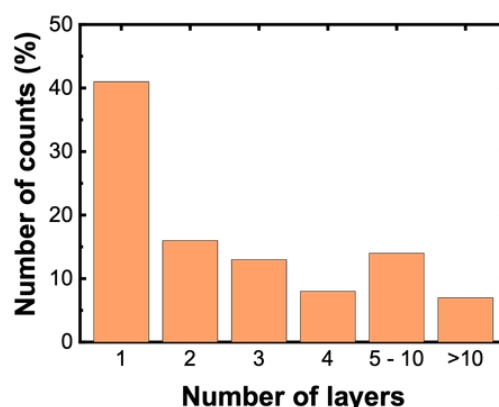


Dynamic light scattering (DLS)

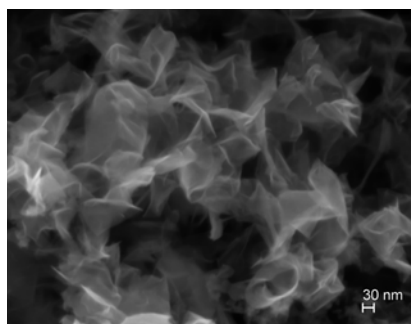
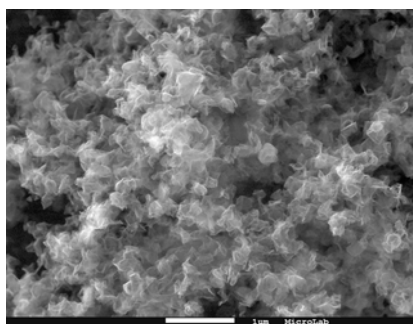
Size distribution of the sheets as obtained by Zetasizer Nano



Percentage of number of atomic monolayers per nanosheets



Scanning Electron Microscopy (SEM)

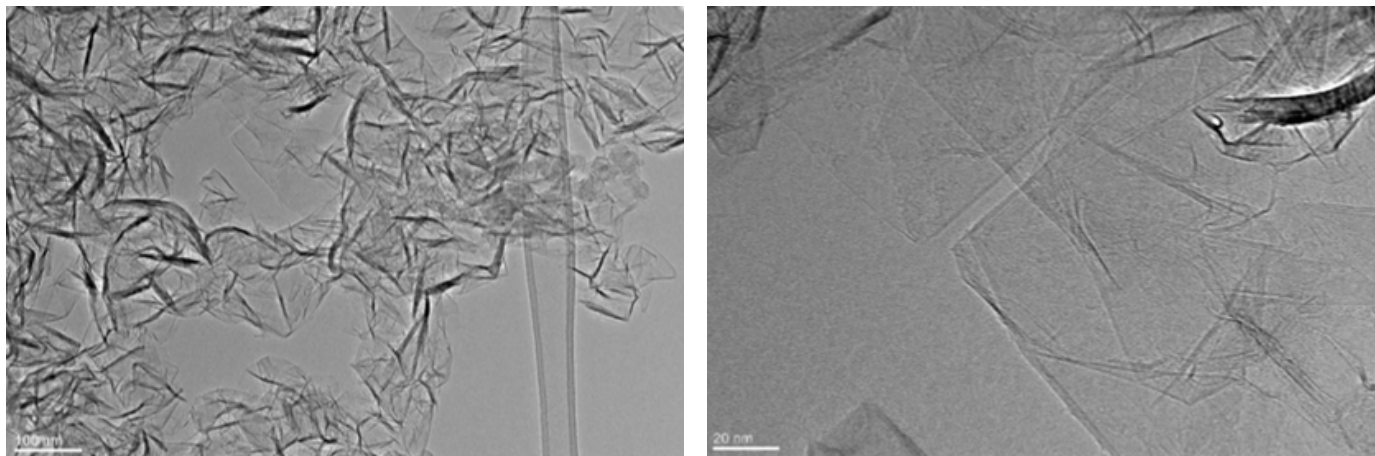


Secondary electron images with different magnification of graphene nanosheets as synthesized.

SEM characterization of the samples has been performed by a JEOL, JSM-7001F field emission gun scanning electron microscope operating in secondary electron imaging mode (SEI) using 15kV accelerating voltage.

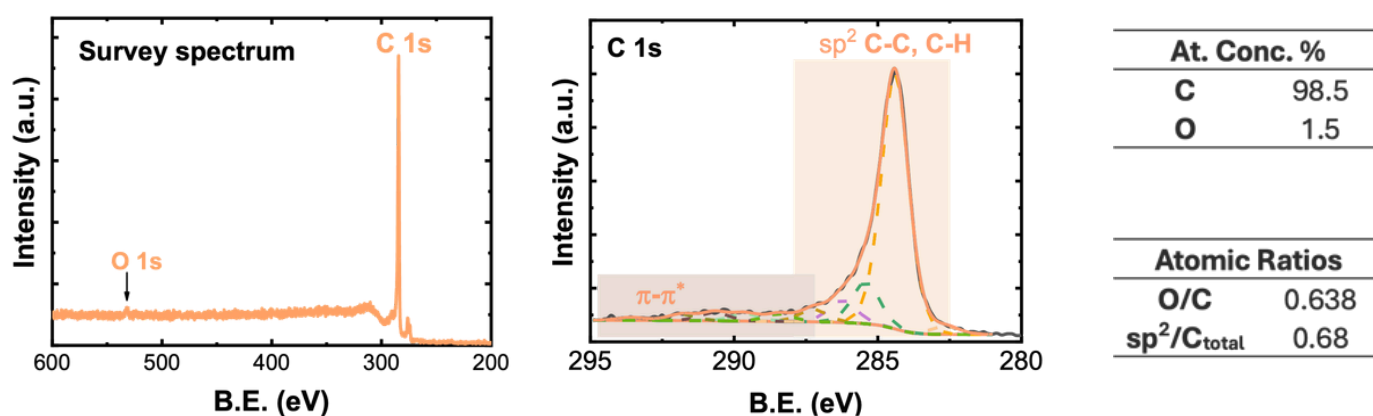
Graphene Characterization & Techniques

High Resolution Transmission Electron Microscopy (HRTEM)



The nanosheets were characterized by a HRTEM JEOL JEM-2010F, operating at 200 kV accelerating voltage.

X-ray Photoelectron Spectroscopy (XPS)



X-ray photoelectron spectra of graphene were obtained with a KRATOS XSAM800 X-ray spectrometer with a double anode (Mg/Al), operating in Fixed Analyzer Transmission mode, with a pass energy of 20 eV and non-monochromatic Mg K α X-radiation (1253.6 eV), produced with a power of 120 W (12 kV x 10 mA).

Unlocking Graphene's Full Potential

▾ About GTechPlasma

We produce customized high-quality graphene-based materials with enhanced properties using an innovative, versatile, cost-effective and environmentally friendly plasma technology.

The technology behind GTechPlasma has been published in a peer-reviewed scientific journal:

Graphene scientific publication
(Reference: Graph_GTP01):

"Plasma-enabled multifunctional platform for gram-scale production of graphene and derivatives"

Applied Materials Today, Vol. 36 (2024),
Article 102056

DOI: 10.1016/j.apmt.2024.102056

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